

Regression with non-linear terms

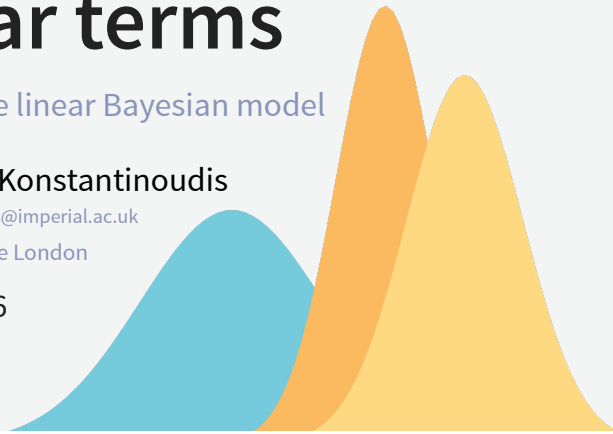
Beyond the linear Bayesian model

Garyfallos Konstantinoudis

g.konstantinoudis@imperial.ac.uk

Imperial College London

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So how do you tell if the relationship is linear?

- Plot each covariate against the response and see what shape the relationship is.
- Make sure you plot it with the transformation used in the model.
- If linear, fit a linear model, if not start thinking of the shape of the relationship.
- Fit as simple a model as possible, do not overcomplicate it unnecessarily. (Occam's razor)

A clarification

In the name normal (or Poisson, binomial, etc.) linear model, the word 'linear' refers to the response being modelled as a linear combination of covariates, i.e.

$$Y_i \sim N(\beta_1 + \beta_2 X_{i2} + \dots + \beta_p X_{ip}, \sigma^2)$$

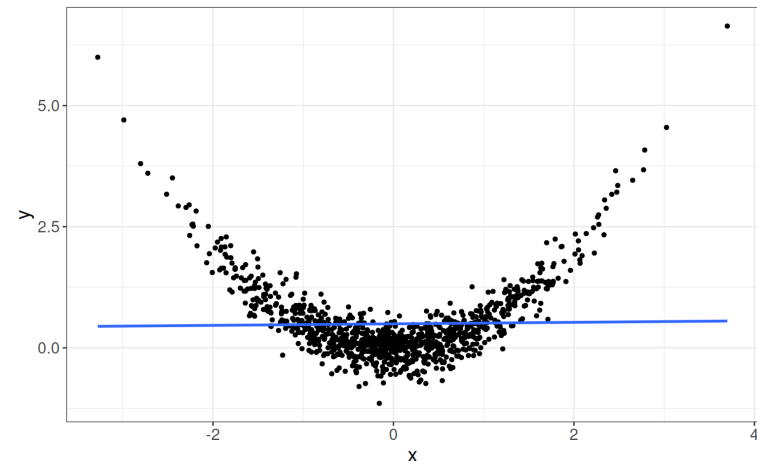
It does not refer to each covariate-response relationship being linear. Therefore the following is also within the class of normal linear models

$$Y_i \sim N(\beta_1 + \beta_2 X_i + \beta_3 X_i^2, \sigma^2)$$

Here, the relationship between Y and x is quadratic, but it is still a linear model.

Example

Associations are not solely linear

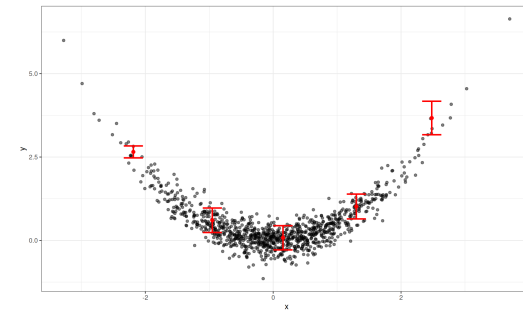


Ways to model non-linearity

- Transform the continuous exposure to categorical variable and model the categories.
- Basis functions (e.g., polynomial, Fourier)
- Natural splines
- Gaussian priors (e.g., random walks, kernels)

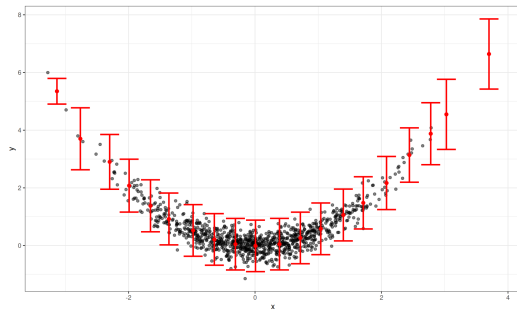
Categorical transformation

- Let's first consider quintiles
- What's the problem with this fit?



Categorical transformation

- More categories...
- What's the problem with this fit?



Basis Expansions

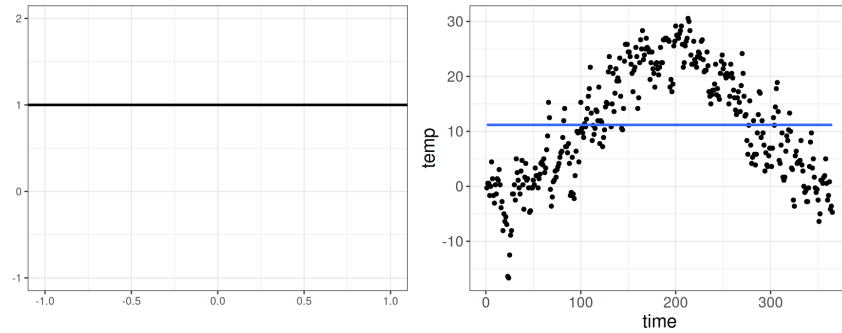
We need to define a set of flexible functions that could capture relationships that are not linear. In general, we can write:

$$Y_i = \sum_j^K \beta_j \phi_j(X_j) + \epsilon_i$$

which we can write: $f(X) = \beta^T \Phi(X)$ and we say that $\Phi(X)$ is a basis system for f .

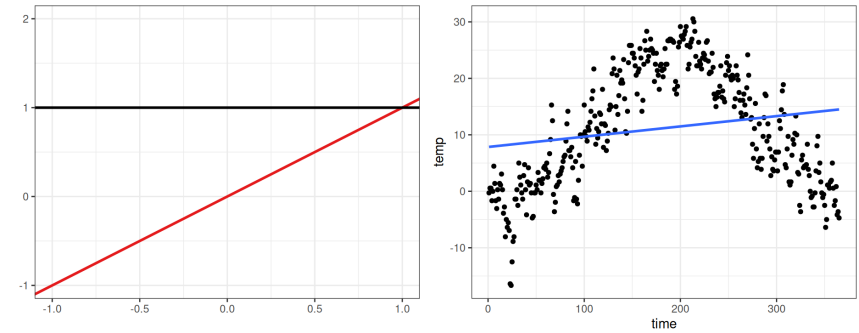
The polynomial basis function

$$\Phi(X) = (1), \text{ thus } Y_i = \beta_0 + \epsilon_i$$



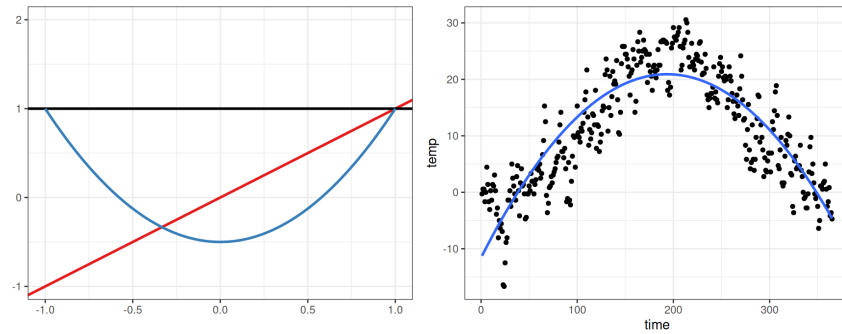
The polynomial basis function

$$\Phi(X) = (1 X), \text{ thus } Y_i = \beta_0 + \beta_1 X_1 + \epsilon_i$$



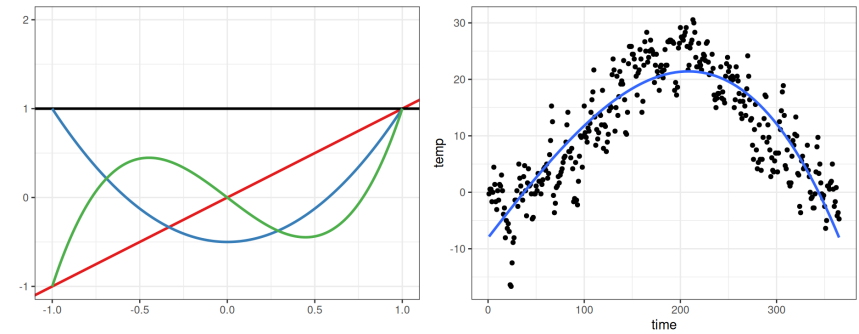
The polynomial basis function

$$\Phi(X) = (1 X X^2), \text{ thus } Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \epsilon_i$$



The polynomial basis function

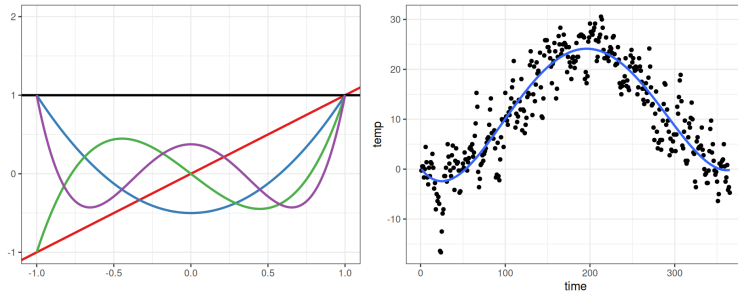
$$\Phi(X) = (1 X X^2 X^3), \text{ thus } Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon_i$$



The polynomial basis function

$$\Phi(X) = (1 \ X \ X^2 \ X^3 \ X^4), \text{ thus}$$

$$Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \epsilon_i$$



Pros and cons

Pros

- These curves are quite flexible—a quadratic can fit most biologically plausible curves
- The curves only use 1(quadratic) or 2(cubic) degrees of freedom more than linear, unlike dummy variable models
- The results are not sensitive to choice of boundaries (there aren't any)
- Outliers mostly influence the extremes of the curve, not the center part

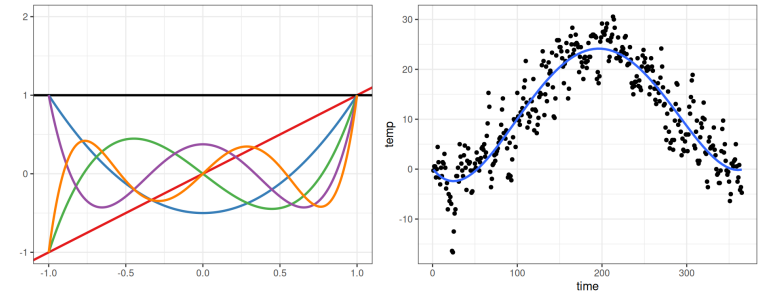
Cons

- They use more degrees of freedom than linear, and therefore have less power
- There is still some sensitivity to influential observations

The polynomial basis function

$$\Phi(X) = (1 \ X \ X^2 \ X^3 \ X^4 \ X^5), \text{ thus}$$

$$Y_i = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \beta_4 X^4 + \beta_5 X^5 + \epsilon_i$$



Other types of basis function

- **Harmonics** - The Fourier basis function

$$1, \sin(\omega X), \cos(\omega X), \sin(2\omega X), \cos(2\omega X), \dots,$$

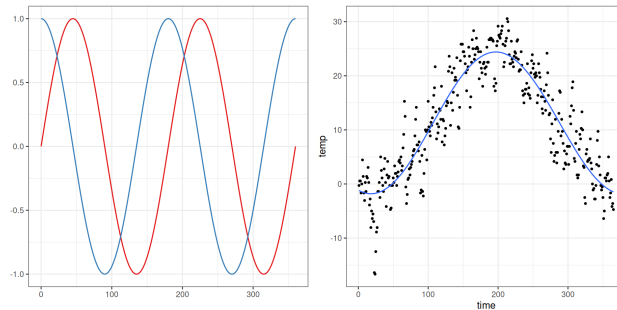
$$\sin(m\omega X), \cos(m\omega X)$$

- constant ω defines the period of oscillation of the first sine/cosine pair. This is $\omega = 2\pi/P$ where P is the period.

Example: Fourier basis function

$$\Phi(X) = (1 \sin(\omega X) \cos(\omega X) \sin(2\omega X) \cos(2\omega X)), \text{ thus}$$

$$Y_i = \beta_0 + \beta_1 \sin(\omega X) + \beta_2 \cos(\omega X) + \beta_3 \sin(2\omega X) + \beta_4 \cos(2\omega X) + \epsilon_i$$



Parametric non-linear effects

- The following models captured the non-linear relationships using simple non-linear functions.
- These types of models are to be preferred if possible, because again they are simpler to make inference from, e.g. the relationship is quadratic.
- However, there may be times when the relationship being modelled does not look like a parametric function. Then what should you do?

Pros and cons

Pros

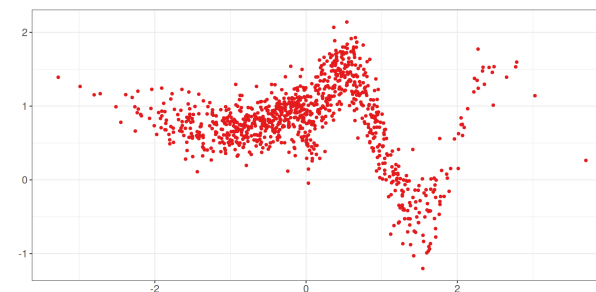
- Excellent computational properties, especially if the observations are equally spaced.
- Natural for describing periodic data, such as the annual weather cycle
- The results are not sensitive to choice of boundaries (there aren't any)

Cons

- functions are periodic; this can be a problem if the data are, for example, growth curves.

Parametric non-linear effects

Consider the following form

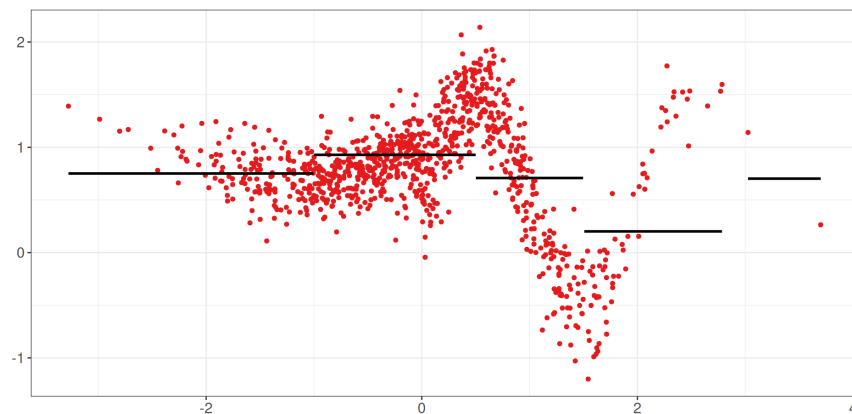


It doesn't look like any simple parametric form (e.g. polynomial, sinusoidal, etc), so what do you do?

Smooth functions

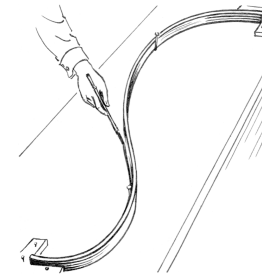
- There are many methods to estimate non-linear relationships such as that on the previous slide.
- They are generically called smooth functions, and include splines (lots of different types), kernel smoothers and local linear smoothers.
- We will focus on splines in this lab, because they are simple to understand graphically and are easy to fit to the data.

Piecewise constant (splines?)



Splines

- Splines were originally thin splints of wood used to trace complex, smooth curves in engineering and architectural.
- Splines were pinned to the drawing at points where the spline changed its curvature.



Piecewise Linear Splines

- We begin fitting splines by dividing the range of exposure into pieces.
- Instead of fitting a constant in each piece, we fit a separate linear term in each piece.
- This has more power since it allows variation within categories to predict variation in outcome.
- We can use fewer categories to capture the deviation from a strictly linear curve because we have slopes within category.

Linear splines

Linear threshold model

$$\begin{aligned} Y_i &= \beta_0^{(1)} + \beta_1^{(1)}x + \epsilon_i & x \leq -1 \\ Y_i &= \beta_0^{(2)} + \beta_1^{(2)}x + \epsilon_i & -1 < x \leq 0.5 \\ Y_i &= \beta_0^{(3)} + \beta_1^{(3)}x + \epsilon_i & 0.5 < x \leq 1.5 \\ Y_i &= \beta_0^{(4)} + \beta_1^{(4)}x + \epsilon_i & 1.5 < x \leq 3 \\ Y_i &= \beta_0^{(5)} + \beta_1^{(5)}x + \epsilon_i & x \geq 3 \end{aligned}$$

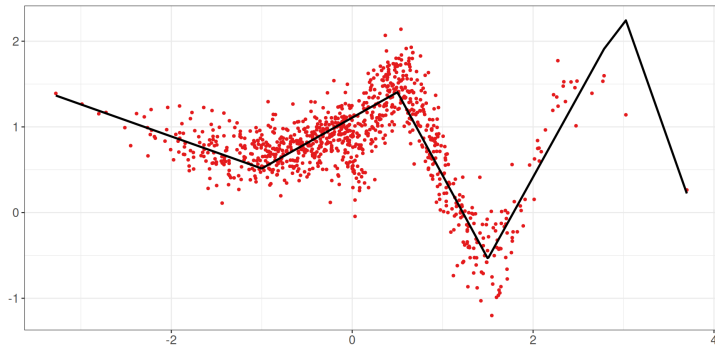
Linear spline model

$$Y_i = \beta_0 + \beta_1 x + \beta_2(x+1)_+ + \beta_3(x-0.5)_+ + \beta_4(x-1.5)_+ + \beta_5(x-3)_+ + \epsilon_i$$

$$(x-k)_+ = \begin{cases} 0, & x < k \\ x-k, & x \geq k \end{cases}$$

Linear splines in R

- And plot the result:



Linear splines in R

- You can use the `bs()` function in R to define the basis function:

```
1 bs(dat$x, degree = 1, knots = c(-1, 0.5, 1.5, 3)) %>%
2   head()
```

```
      1      2      3 4 5
[1,] 0.7273541 0.27264593 0.0000000 0 0
[2,] 0.3156038 0.68439625 0.0000000 0 0
[3,] 0.7731419 0.00000000 0.0000000 0 0
[4,] 0.8407311 0.00000000 0.0000000 0 0
[5,] 0.0000000 0.32151084 0.6784892 0 0
[6,] 0.9561009 0.04389912 0.0000000 0 0
```

Cubic splines

Similarly we can define higher order polynomial splines, for instance:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \\ &\quad \beta_4(X+1)_+ + \beta_5(X+1)_+^2 + \beta_6(X+1)_+^3 + \\ &\quad \beta_7(x-0.5)_+ + \beta_8(x-0.5)_+^2 + \beta_9(x-0.5)_+^3 + \\ &\quad \dots + \epsilon_i \\ (x-k)_+ &= \begin{cases} 0, & x < k \\ x-k, & x \geq k \end{cases} \end{aligned}$$

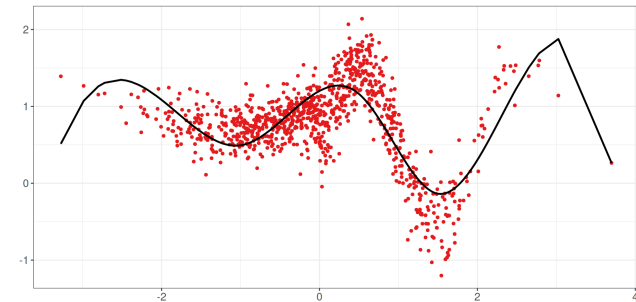
which reduces to the following to ensure smooth curvature on the knots (it can be seen after deriving the first and second derivative):

$$Y_i = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_6(x+1)_+^2 + \beta_9(x-0.5)_+^2 + \dots + \epsilon_i$$

Cubic splines

- Similar as before:
- `bs(x, degree = 3, knots = c(-1, 0.5, 1.5, 3))`:

without the splines package:



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Splines

A spline of order n is a piecewise polynomial function of degree $n-1$ in a variable x .

(Basis) Splines can be:

- Piecewise constant.
- Linear.
- Quadratic.
- Cubic.
- higher order polynomials.
- etc.

Why are the borders so wiggly?

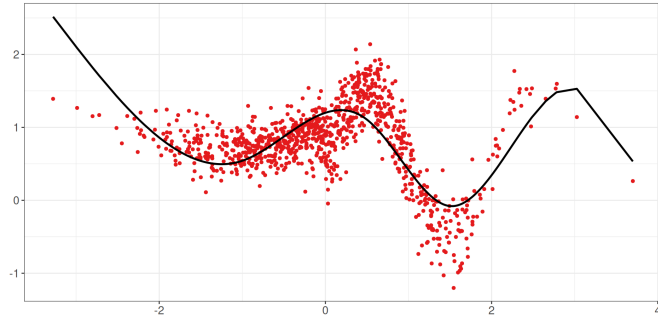
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Natural cubic splines

- Splines can have high variance in the boundaries.
- This problem aggravates if knots too few.
- The natural spline constraints the fit to be linear before and after the first and last knots.
- This stabilises the fitting and makes a more reasonable assumption.

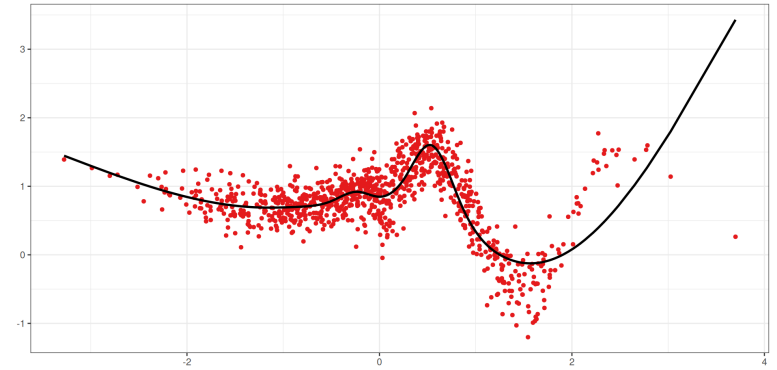
Natural cubic splines in R

- To define natural cubic splines in R:
- `ns(x, knots = c(-1, 0.5, 1.5, 3))`:

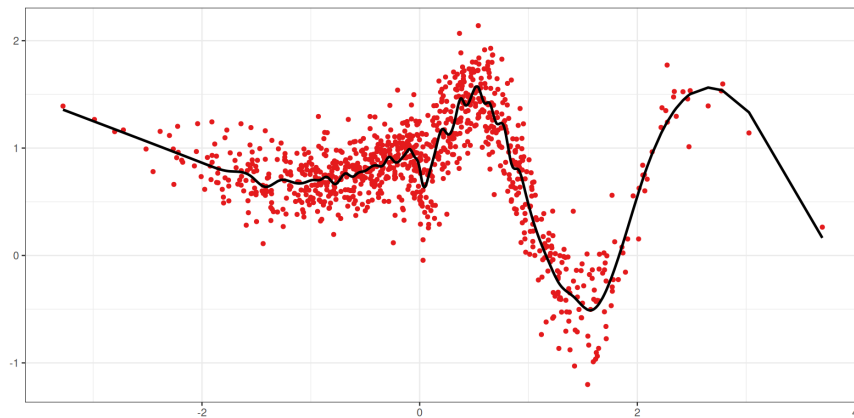


Well, better fit be achieved with more knots

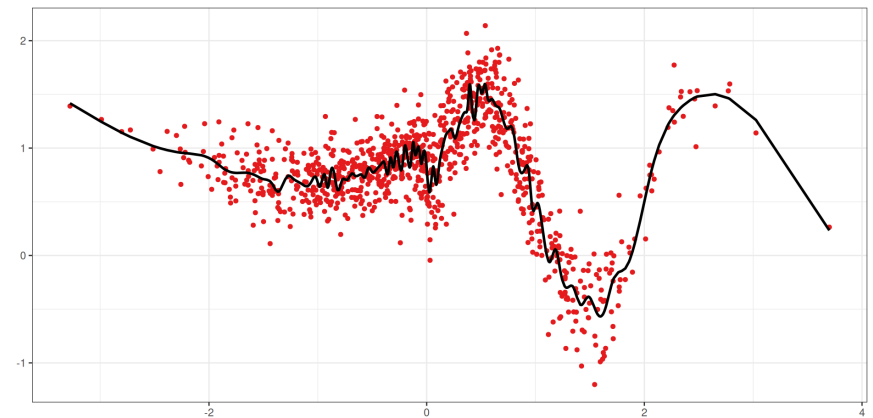
- `ns(x, df = 10)`



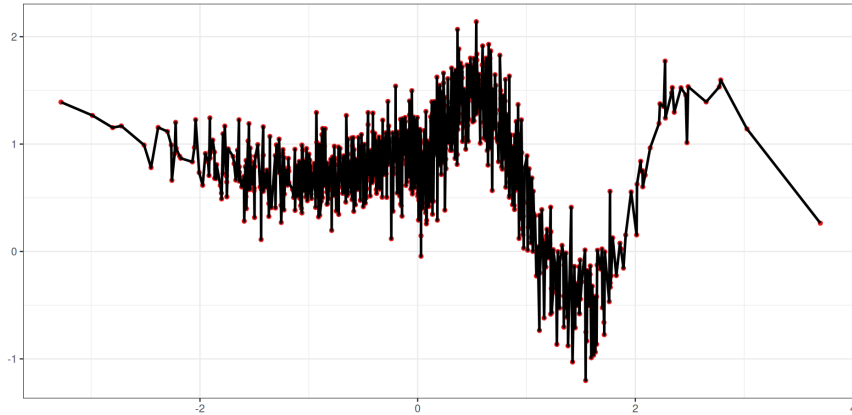
... and more



... and more



... and more



is this a useful model?

Gaussian priors

- Random walks
- Autoregressive processes
- Gaussian kernels (out of the scope)
 - Correlation is a function of distance
 - “Continuous domain models”
 - Exponential, Gaussian, Matern, etc. (correlation functions)

Random walk of order n

- Random walk of order 1

$$x_i = x_{i-1} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

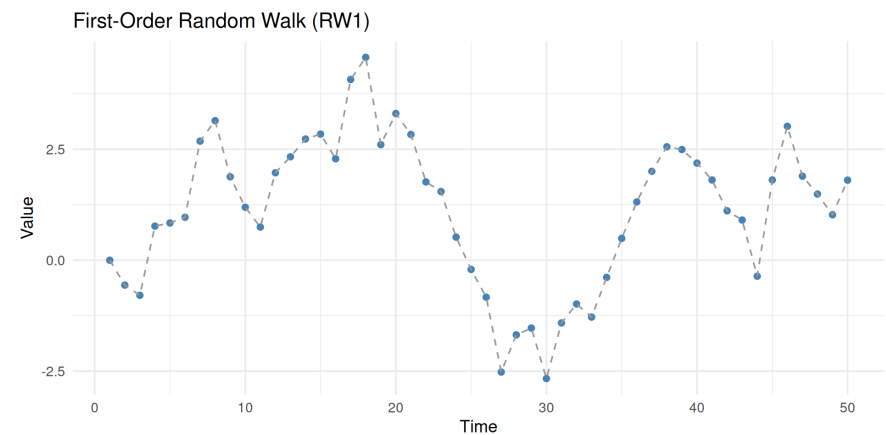
- Random walk of order 2

$$x_i = 2x_{i-1} - x_{i-2} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

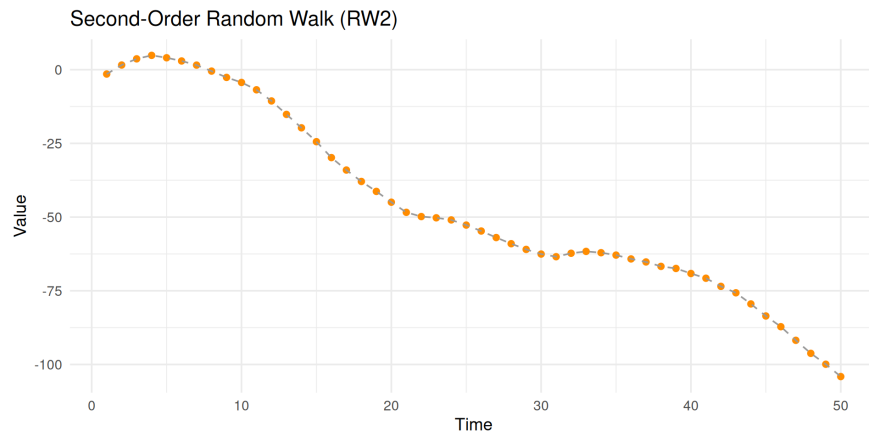
- This is very similar with the categorical transformation example, but more concise.

The prior controls smoothness: the stronger it is the smoother the result. (given the data)

Random walk of order 1 example

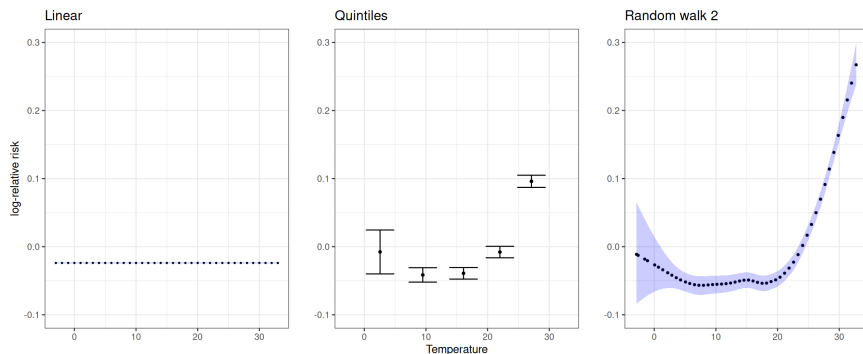


Random walk of order 2 example



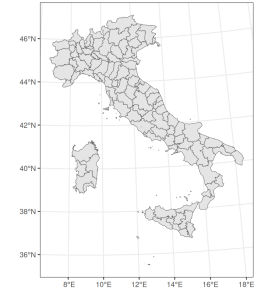
Example

- Fit a linear, a categorical and a random walk model for capturing the effect of temperature.



Example: Mortality and temperature in Italy

- 1,946,755 summer deaths during 2011-2018 in Italy
- Data is available at 107 administrative units
- Temperature is available from ERA-5 at 9km resolution



Autoregressive process of order n

- Autoregressive process of order 1:

$$x_i = \phi x_{i-1} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

- Autoregressive process of order 2:

$$x_i = \phi_1 x_{i-1} + \phi_2 x_{i-2} + \epsilon_i, \quad \epsilon_i \sim N(0, \tau^{-1})$$

- when $\phi = 0$ we have RW1 or RW2
- To complete the parametrisation, we need to define priors of ϕ . Can you think of a prior?

Extensions

- It is straight forward to include confounders in the linear predictor.
- Confounders can be included as linear or more flexible terms.
- We can account for spatiotemporal correlations.
- We can incorporate the delayed effect of temperature on health (lags)
 - joint lag models
 - distributed non-linear lag models

Summary

- Introduction to non-linear exposure models
- Theory and application of linear splines
- Understand more flexible non-parametric functions (focus on splines)
- Random walk of order 2 to model non-linearity

Questions?